

YARN (EXERCISE)

Tuesday, July 01, 2025 9:18 PM

4.1 What is YARN?

1. In your own words, explain why YARN is often called the “operating system” of a Hadoop cluster.
2. Name two non-MapReduce engines that can run on YARN and describe one benefit of this multi-engine support.

4.2 YARN Architecture

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2. Outline the sequence of steps—from client submission to container launch—that takes a job through YARN’s execution flow.

4.3 YARN Scheduler Types

1. Briefly contrast the FIFO, Capacity, and Fair schedulers in terms of resource allocation policies.
2. Give one scenario where you’d prefer the Capacity Scheduler over Fair Scheduler, and explain why.

4.4 YARN Responsibilities

1. List three key responsibilities YARN handles beyond simply launching containers.
2. How does YARN enable different frameworks (e.g. Spark, Tez, MapReduce) to coexist on the same cluster?

4.5 Fault Tolerance in YARN

1. Describe what happens when a NodeManager fails during a running application.
2. Explain how YARN achieves ResourceManager high availability in Hadoop 3+.

4.6 Real-World Use Case

1. If you submit Spark, Tez, and MapReduce jobs simultaneously, how does YARN isolate and manage their resource requests?
2. What role does the ApplicationMaster play when coordinating a multi-stage job like Spark’s shuffle?

4.7 YARN CLI Commands

1. Which yarn command lists all running applications, and how would you filter it to show only your jobs?
2. How do you kill a misbehaving YARN application from the command line?
3. Where can you go in your browser to view YARN’s web UI and monitor cluster usage?

4.8 Why Performance Tuning Matters

1. Name two risks of running a poorly tuned Hadoop cluster in production.
2. Pick one tuning area (Memory & CPU, Disk I/O, Network, or Config files) and describe one concrete change you might make.

4.9 HDFS Tuning

1. How would increasing the HDFS block size from 128 MB to 256 MB affect small-file versus large-file workloads?
2. Which configuration property controls the default replication factor, and when might you lower it?
3. Describe two strategies for mitigating the “small files” problem on HDFS.